

ICP-BASE

Common Pitfalls

Sebastian Stumpf

sebastian.stumpf@unibe.ch

This guide is made to help users troubleshoot common errors and unexpected behavior during data processing. Please try fixing the problem first using this guide. If you cannot identify and resolve the problem, feel free to write me an email! Trust me, I wished as much as you that you didn't need it but here we go:

When reporting an issue, please include the following:

- The ICP-Base_log.txt (found in Documents/ICP-Base on your computer)
- A detailed description of the problem
- The saved base, project, or session file (.mat)
 - All data shared with me is treated as strictly confidential and will not be passed on to any third party. And frankly, I have neither the time nor the inclination to publish your data before you — I'm already drowning in my own.
- The brand and model of your LA and ICP-MS system
- If the issue occurs during import: the exact raw data files that failed to load and the brand and model of your LA and ICP-MS system from which the file originates
- Which operating system you are using

1. Base

- Why does my base not show any projects or sessions?
 - Did you load a base, project or session (.mat) file that you previously saved somewhere? If no there is no data to show. If you did not save your work previously it is gone.
- Why am I not able to export or visualize my data via the Table Builder Submodule?
 - Did you do the analyte-calibrator-assignment in the Results Module output table configuration Tab? This is essential for building the tables.

2. Import Module

- Why does my import fail?
 - Do your file type match the LA or ICP-MS brand and layout that you actually selected in the dropdown menus? Make sure to really match this.
 - Did you tinker with the files beforehand? This often breaks the format and the import fails accordingly.
 - If you used templates (or for any metadata), make sure to always match the number of spots. If you imported e.g. 50 spots and your template (or any metadata) contains e.g. 60 entries, this will fail.
 - All data must contain the same elements. No mix-and-match with files holding different element cps data.
 - If thoroughly looked at and made sense of what you are trying to import and it still does not work or you really have to tinker with the files write me an email. Maybe the LA or ICP-MS data is a different version which is slightly different from the version ICP-Base was programmed and tested on.
- My Acquisition Times are wrong
 - Did you tinker with the files beforehand? For some data types the acquisition time is the time the file was created. If you copied the file or altered and saved it again, this time might have changed and is not representative anymore. See Extensive Manual for more information on this.
 - If you imported them via the template make sure to match the spot number and to use the exact same format as in the template.
 - If none of this is the case please write me an email, there might be problem with 12h and 24h formats.

3. Interval Module

- Why are my intervals not properly visualized?
 - Did you correctly detect the signal starts and ends in the Import Module? If no the visualization will not work properly. You can jump back to the Import Module and change the detection threshold for the Single file type and use the global indent feature for the continuous file type to detect them properly. See the extensive manual for more information.
- Why does the Zoom in to Signal Checkbox zoom somewhere non-sensical?
 - Did you correctly detect the signal starts and ends in the Import Module? If no the visualization will not work properly. You can jump back to the Import Module and change the detection threshold for the Single file type and use the global indent feature for the continuous file type to detect them properly. See the extensive manual for more information.
- Why does the global interval setting feature set my intervals somewhere non-sensical?
 - Did you correctly detect the signal starts and ends in the Import Module? If no the visualization will not work properly. You can jump back to the Import Module and change the detection threshold for the Single file type and use the global indent feature for the continuous file type to detect them properly. See the extensive manual for more information.
- Why can't I move the intervals anymore?
 - Is the Datatips On/Off toggle button activated? If yes deactivate it (bottom right).
- Why does my background corrected data still show data for the background?
 - Because this works with averages and what you see is the scatter above the mean. Don't worry about it.
- Reference materials and everything that is not an inclusion needs "Matrix" intervals and not "Inclusion" intervals. This could impact e.g. the drift correction and the quantification.

4. Classification Module

- Why am I not seeing any RMs in the listbox?
 - Did you press the Get RM files button? Did you set the path to the RM folder properly (if you changed it?). Are does the Documents/ICP-Base/RM_Files

folder exist and have .csv files? Are the .csv files in the correct format (Same as the default ones)?

- You can right-click on a green line (so CAL and Type already chosen) in the table to check the RM values.

5. Drift Module

- If you see an error here it means something has gone wrong in the modules before!
 - Do your elements have a name like Li7, Na23, La139? This is the correct naming for the software, it will break if your element names are different. If this is the case there is a problem at the Import that is probably not even your fault (unless you tinkered with the files), contact me.
 - Did you correctly do the classification (CALs, Bracket IDs etc)? ICP-Base only supports bracketing (e.g. CAL, CAL, UNK, UNK, CAL, CAL), there is no step-forward (e.g. CAL, CAL, UNK, UNK, UNK...)! If your data is not bracketed by the standards (e.g. they were analyzed only at the start of the session but not the end) you're out of luck. Read the extensive manual if you're confused.
 - Did your RM files hold the element data and are in the correct format? If no this might be it.
 - Did you set "Matrix" intervals for all calibrators? If the spots are classified as calibrators but they have no intervals (or "Inclusion" intervals) the drift correction cannot use them and fails. Either give them intervals or set them to Unknown

6. Quantification Module

- Why am I not seeing certain spots?
 - They might be outside your current CAL-bracket ID bracket. Change one of them in the dropdowns to see the other brackets.
- Why does the string compare not work?
 - Did you select the correct column to compare to?
- Why is the calculation failing?
 - Check if the software correctly imported the oxides via Settings > Show Oxides that will be used. If this does not work there is a problem with your Oxides.csv which should be found in Documents/ICP-Base/Total_Oxide_Calculations.

- The calculation generally only fails if something in the Modules beforehand is wrong. The quantification is just the culmination of everything that was done beforehand.

7. Results Module

- Why am I not seeing anything in the QCRM plots?
 - Did you set a QCRM that is in the current bracket? Check your CAL and bracket id
 - Check your RM files where the RM data is stored – also check the format of the RM file.